

Software Requirements Specification (SRS)

Project Title

3D Print Sales Tracker

Project Overview

The 3D Print Sales Tracker is a web application designed to help small business owners keep track of their sales in a simple and organized way. The idea for this project came from my own experience running a small 3D printing business. I previously tracked sales using Google Forms and spreadsheets, but the process became time-consuming and difficult to manage as more sales were recorded.

This application is intended to make sales tracking faster and easier by allowing users to enter sales through a mobile-friendly interface, view revenue data, and customize the sales form to fit their business needs.

Purpose

The purpose of this project is to create a tool that helps small business owners record and organize sales without relying on multiple spreadsheets or forms. The application focuses on making data entry quick and easy while still providing useful information about revenue and product performance.

Functional Requirements

Navigation

The system will provide navigation between the following pages:

- Home
- Sales
- Revenue
- Settings

Sales Page

The Sales page will allow users to record a sale by completing a step-by-step process.

Users will be able to:

- Select one or more products sold in an order.
- Assign quantities to each product using on-screen controls.
- Assign colors to each individual print sold.
- Enter the total purchase amount for the order.
- Select where the customer found the product.
- Save the sale to the system.

Product Selection

Products are displayed as selectable cards instead of a traditional dropdown menu.

The system will:

- Allow multiple products to be selected.
- Allow users to increase or decrease quantities for each product.
- Support product images in future versions.

Color Selection

The system automatically creates color selections based on the products chosen.

For example, if a customer purchases two Butterfly Knives, the system will generate:

- Butterfly Knife #1 Color
- Butterfly Knife #2 Color

This allows each print in an order to have its own color assignment.

Revenue Page

The Revenue page displays detailed sales analytics and reporting information.

Users can:

- View total revenue earned.
- View sales timestamps and order history.
- View sales grouped by month.
- Open a monthly report showing all sales for a selected month.
- Generate a printable monthly sales report.
- Filter sales by date range.
- View product performance charts.

- View color usage charts.
- View sales source charts showing where customers found products.

Sales History

The system stores a timestamp for every sale and displays sales in a paginated table.

The system will:

- Display up to 10 sales per page.
- Provide Previous and Next page navigation.
- Support horizontal scrolling on mobile devices.
- Display sales in chronological order.

Settings Page

The Settings page functions as a form builder that allows users to customize the sales form.

Users can:

- Edit question text.
- Change question types.
- Reorder questions.
- Add new questions.
- Delete questions.
- Add answer choices.
- Remove answer choices.
- Modify existing answer choices.

Supported question types include:

- Dropdown
- Multi Image Choice
- Money Keypad
- Number Input

Data Storage

The application now stores data using Supabase cloud storage.

The system will:

- Save sales data to a cloud database.
 - Sync sales across multiple devices.
 - Retrieve sales information when the application loads.
 - Store timestamps and reporting data online.
 - Support future user account integration.
 - Usability
 - The application should be easy to use on desktop and mobile devices.
 - Navigation should remain usable on small screens.
 - Tables should support horizontal scrolling when needed.
 - Sales confirmation messages should provide clear feedback after submission.
 - Delete actions should require user confirmation before removing data.
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Non-Functional Requirements

Performance

- The application should load quickly.
- Sales should save immediately after submission.

Usability

- The application should be easy to use on both desktop and mobile devices.
- Buttons and controls should be large enough for touch screens.
- Users should be able to record sales with minimal typing.

Reliability

- Sales data should remain available after page refreshes.
 - Saved settings should remain available between sessions.
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Future Improvements

Some features planned for future development include:

- User accounts and authentication
- Row Level Security (RLS) for user data isolation
- PDF export improvements
- Inventory tracking
- Expense tracking
- Profit calculations
- CSV and Excel exports

- Etsy integration
 - Multi-user support
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Changelog

Original Project Idea

- Planned as a Windows Forms desktop application.
- Included basic sales tracking features.

Web Application Conversion

- Switched from Windows Forms to a web-based application using HTML, CSS, and JavaScript.
- Added a navigation system.
- Added Home, Sales, Revenue, and Settings pages.
- Added local storage support for saving data.

Form Customization Update

- Added dynamic form generation.
- Added editable questions and answer choices.
- Added a dedicated Settings page for customization.

Current Progress

Current Progress

- Added multi-select product cards.
- Added animated quantity selectors.
- Added dynamic color questions based on selected products.
- Added individual color assignments for every print sold.
- Added order-based purchase totals using a custom keypad.
- Added customizable question text and answer choices.
- Added a form builder interface for managing sales form questions.
- Added support for multiple question types.
- Added question reordering functionality.
- Added question creation and deletion functionality.
- Added mobile-friendly navigation improvements.
- Added responsive table scrolling.
- Added pagination to sales history.
- Added pagination to revenue timestamps.
- Added monthly sales grouping and reporting.

- Added product performance charts.
- Added color usage charts.
- Added color-by-product analytics.
- Added sales source pie charts.
- Added monthly report generation.
- Added sale submission confirmation screens.
- Added delete confirmation prompts.
- Integrated Supabase cloud database storage.
- Deployed the application using GitHub Pages.
- Improved mobile responsiveness throughout the application.